Differentiable 3-Axis CNC Machining: Co-Optimizing Toolpaths and Feedrates via Smooth-to-Hard Surrogate Gradients and Residual Reinforcement Learning

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Abstract—Generating physically optimal toolpaths for 3-axis Computer Numerical Control (CNC) milling requires reasoning over long-horizon trajectories in which every cutting pass permanently mutates the in-process workpiece (IPW) geometry and governs subsequent cutter-workpiece engagement (CWE) loads. Classical computer-aided manufacturing (CAM) relies on conservative geometric offset heuristics and constant feedrates because Boolean solid-modeling kernels are discontinuous—yielding zero gradients during air-cutting and undefined derivatives at stock boundaries—while naive geometry-only optimization under exact-stop kinematics fails to exploit real CNC corner blending or respect volumetric chip-load limits. We present DiffCAM, an MIT-licensed, GPU-accelerated dual-mode differentiable 3-axis milling framework that couples exact Boolean swept-cylinder signed distance field (SDF) carving with a log-sum-exp smooth relaxation, Straight-Through Estimator (STE) stiffness continuation, acceleration-bounded junction corner blending aligned with NIST RS274 (G64 P0.05) and Haas Next-Gen Control (G187 P2 E0.05), and an analytical Peak Material Removal Rate (MRR) barrier. Seeded by modular clean-room CAM engines spanning Facing, Pocket, trochoidal Adaptive, and Contour operations (37–370 segments) and parameterized via a drift-free, wall-clearance-weighted JointPathFeedSpline, we conduct a controlled study comparing first-order differentiable trajectory and policy optimization (STE-Continuation-Adam, Annealed-Adam, DiffPolicy-SHAC) against zero-order reinforcement learning and black-box search (Hard-PPO, Hard-CEM, Hard-CMA-ES) across 16 tasks under a strict 64-call simulator budget. Smooth-to-hard continuation (STE-Continuation-Adam and Annealed-Adam) reduces mean cycle time by $\mathbf{42.0 \pm 13.5}$% and $\mathbf{43.0 \pm 15.7}$% (122.2 s $\rightarrow$ 73.9 s) at **100**% hard feasibility—nearly 4$\times$ the gain of evolutionary search (9.3%–11.3%)—while first-order DiffPolicy-SHAC achieves 2.0$\times$ the speedup of zero-order Hard-PPO (**37.0**% vs. **18.5**%) and leads all methods on enclosed Pocket roughing (**32.1**%). Ablations show that removing our volumetric MRR barrier triggers +136.7% cutter overload, and independent verification across 6 industrial CNC dialects (48/48 pass), a 0.125 mm posted G-code evaluator, and containerized CAMotics v1.2.0 certifies 8/8 watertight STL solids with 0.00 mm^3 gouge, 0.00 mm^3 uncut residual, and interactive 3D stock-and-toolpath simulation bundles.

I. Introduction

Subtractive CNC machining remains the backbone of precision mechanical manufacturing, yet toolpath generation is still dominated by purely geometric rules devised decades ago [1, 7]. Unlike collision-free robot motion planning in static environments, 3-axis milling is a history-dependent, irreversible manipulation process over a continuous volumetric field: every millimeter traveled by the rotating endmill permanently carves the in-process workpiece (IPW), determining the radial and axial cutter-workpiece engagement (CWE) encountered by future passes [12, 18]. Consequently, standard industrial CAM packages decouple toolpath topology (e.g., zig-zag facing, offset pocketing, trochoidal clearing, exterior contouring) from conservative, worst-case constant feedrates chosen to survive full-width slotting corners—leaving substantial machine productivity unrealized during partial-engagement cuts and air-repositioning transitions.

Bridging modern gradient-based trajectory optimization and reinforcement learning (RL) with CNC milling faces three coupled obstacles:

1) Discontinuous Boolean Material Removal: In solid modeling and dexel/voxel kernels [3, 19], subtracting a swept tool volume ψ_{tool} from stock SDF ϕ_t via $\phi_{t+1}(\mathbf{x}) = \max(\phi_t(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$ yields identically zero spatial gradients when a segment cuts air and non-differentiable kinks at cutter engagement boundaries [16].
2) Surrogate Bias and Cumulative Drift: Naive smooth relaxations of $\max(\cdot, \cdot)$ introduce volumetric occupancy inflation at low stiffness k, causing optimizers to hallucinate phantom material removal that fails exact Boolean evaluation [11, 17]. Simultaneously, parameterizing multi-pass toolpaths via relative segment deltas $\Delta\mathbf{p}_t$ induces $O(T)$ cumulative position variance across $T \sim$ 37–370 segments, slamming interior passes into blueprint walls.
3) Controller-Blind Kinematics: Optimizing path length under constant feedrates and exact-stop (G61) assumptions stalls when part tolerances fix the required pass count, whereas naively maximizing feedrates without volumetric engagement feedback causes catastrophic cutter overload ($\text{MRR}_t \gg \text{MRR}_{\max}$) or mismatches physical CNC corner-rounding deceleration [4, 10].

These challenges motivate a central question in differen-

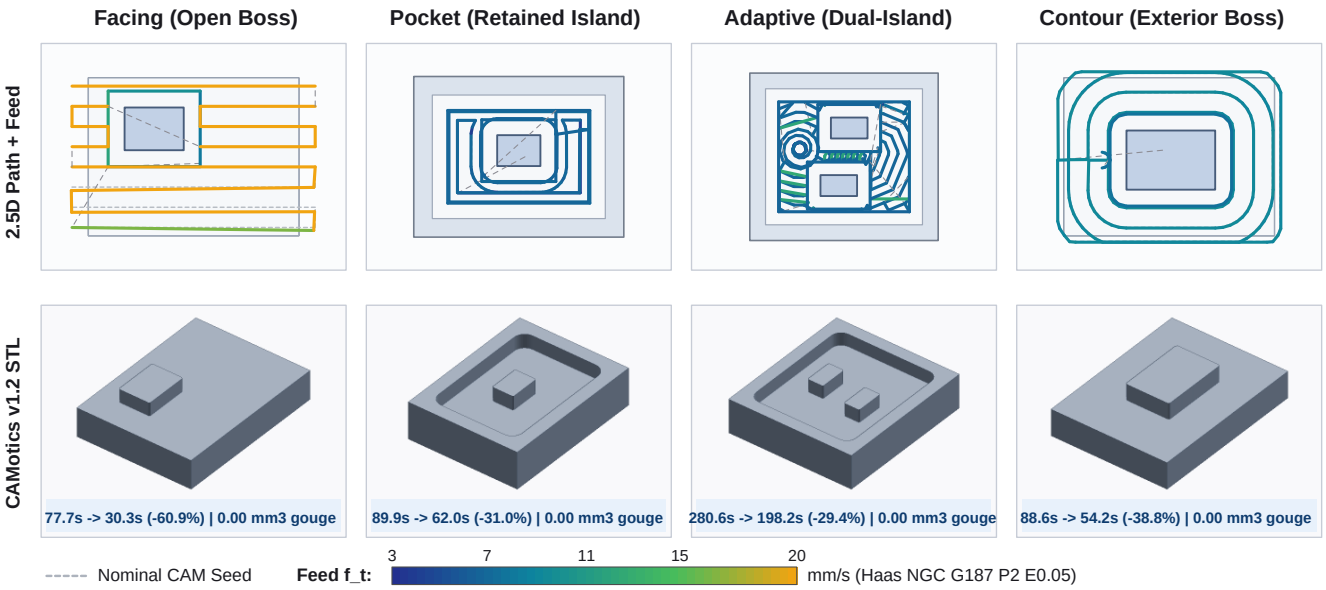


Fig. 1. DiffCAM joint 2.5D toolpath and feedrate co-optimization certified in CAMotics v1.2.0. Top row: Representative held-out test parts across four modular MIT clean-room CAM strategy families showing stock billets, retained islands/bosses, dashed gray nominal CAM seeds (37–370 segments), and optimized STE-Continuation-Adam trajectories colored by per-segment feedrate $f_t \in [2.8, 20.0]$ mm/s under Haas Next-Gen Control corner blending (G54, G187 P2 E0.05). Bottom row: 4×-MSAA isometric renderings of the closed, watertight binary STL meshes reconstructed directly by containerized CAMotics v1.2.0 (0.25 mm grid) from posted Haas NGC G-code, achieving 29.4%–60.9% cycle-time reductions on these four parts (39.5% mean across all 8 test parts) with 0.00 mm^3 target gouge, 0.00 mm^3 residual, and exact STL volume identity with nominal CAM.

tiable physical AI: when initialized from valid topological CAM seeds under a tight simulation budget, how do first-order analytical gradients through a smoothed-to-hard cutting simulator compare against zero-order RL and evolutionary strategies operating directly on exact Boolean cutting?

To answer this question rigorously, we introduce DiffCAM (Fig. 1), an end-to-end differentiable 3-axis milling simulator, trajectory optimizer, and residual RL benchmark built in Taichi [8]. Our primary contributions are fourfold:

- Dual-Mode Swept-CSG Simulation with STE Continuation: We formulate a GPU-accelerated 3D SDF simulator supporting both exact Boolean material removal and a log-sum-exp smooth surrogate, paired with a Straight-Through Estimator (STE) [2] stiffness schedule (k : 25 $\rightarrow$ 90) and exact hard-feasibility checkpointing that eliminates soft-to-hard occupancy drift.
- Blended CNC Kinematics and Volumetric MRR Barrier: We co-optimize continuous 2.5D XY toolpaths and per-segment feedrates under a differentiable acceleration-bounded junction blending model aligned with RS274 G64 P0.05 and Haas Next-Gen Control G187 P2 E0.05, constrained by differentiable pre-cut chip-volume Peak MRR, analytical wall-gouge, holder-collision, and rapid-clearance barriers.
- Clean-Room CAM Engines, Drift-Free Co-Parameterization, and Controlled RL Study: We build modular MIT clean-room CAM engines (facing, pocket, trochoidal adaptive, contour), introduce an analytical wall-clearance-weighted JointPathFeedSpline that eliminates $O(T)$ cumulative delta drift while pinning plunge/retract and finishing boundaries, and compare first-order STE-Continuation-Adam, Annealed-Adam, and DiffPolicy-SHAC [20] against zero-order Hard-PPO [15], Hard-CEM [13], and Hard-CMA-ES [6] under strict simulator-call parity.
- Multi-Controller G-Code, CAMotics STL, and Interactive 3D Simulation: Every held-out test solution is post-processed across 6 industrial CNC dialects (RS274, Haas NGC O01001–O01008, FANUC, Robodrill, Makino, Mazak), exported to native .rustcnc, .camotics, and WebGL 3D stock-and-toolpath simulators, and certified by both a 0.125 mm-step evaluator and containerized CAMotics v1.2.0 [3] (48/48 dialect pass, 8/8 watertight STLs, 0.00 mm^3 gouge, 0.00 mm^3 residual, and 36.2% aggregate / 39.5% mean test cycle-time reduction).

II. Related Work

Geometric CAM and Engagement-Aware Toolpaths. Classical pocket and contour milling rely on 2D Voronoi medial-axis offsets [7] and direction-parallel rastering, which suffer from sharp radial engagement spikes in concave corners. Trochoidal and adaptive clearing strategies [1, 12] bound radial immersion via circular or cy-

cloidal loops, while solid verification engines use Z-buffer dexels [19], R-functions [16], or octree marching-cubes isosurfaces as in CAMotics [3]. However, these forward geometric kernels are non-differentiable, preventing direct gradient-based co-adaptation of waypoint geometry and feed schedules to part-specific stock state.

CNC Corner Blending and Feedrate Scheduling. Physical CNC interpolators avoid coming to a complete stop at every linear G1 block junction (G61) by blending adjacent velocity vectors within a user-specified path-deviation tolerance—standardized as G64 Pe in NIST RS274NGC [10] and G187 P2 Ee on Haas Next-Gen Control [4, 18]. Classical feedrate scheduling optimizes speed along a frozen reference path under axis acceleration or cutting-force bounds [1, 4]. Conversely, optimizing path geometry under frozen feeds yields negligible speedup when feature walls lock pass topology. DiffCAM unifies both regimes by backpropagating simultaneously through blended junction kinematics, swept chip-volume engagement, and waypoint positions.

Differentiable Physics vs. Zero-Order RL. Differentiable simulators such as DiffTaichi [8] and Brax [5] enable low-variance first-order trajectory optimization and short-horizon analytic actor-critic training (SHAC) [20]. Yet Suh et al. [17] and Metz et al. [11] showed that stiff contact discontinuities can induce high gradient variance or surrogate bias, allowing randomized zero-order estimators such as PPO [14, 15], CEM [13], and CMA-ES [6] to remain competitive. By combining log-sum-exp CSG smoothing with Straight-Through Estimator (STE) evaluation [2] and stiffness continuation, DiffCAM preserves high gradient signal-to-noise ratio (SNR $\in [30.5, 62.2]$) across 37–370 cutting segments while guaranteeing zero surrogate gap at deployment.

III. Differentiable 3-Axis Milling Formulation

A. Dual-Mode Swept-Cylinder SDF Material Removal

We represent the 3D stock billet on a uniform Eulerian grid $\Omega \subset \mathbb{R}^3$ of voxel pitch Δx with signed distance values $\phi_t : \Omega \to \mathbb{R}$ after segment $t \in \{0, \dots, T\}$, following the convention $\phi_t(\mathbf{x}) < 0$ inside solid material and $\phi_t(\mathbf{x}) > 0$ in empty space. Given a flat-end cylindrical cutter of radius R, flute length L_{flute}, holder clearance height L_{clear}, and holder shank radius $R_{\text{holder}} > R$ moving linearly from $\mathbf{p}_{t-1} \in \mathbb{R}^3$ to $\mathbf{p}_t \in \mathbb{R}^3$, let $\tau^\star(\mathbf{x}) = \text{clip}\big(\frac{(\mathbf{x}-\mathbf{p}_{t-1})\cdot\Delta\mathbf{p}_t}{\|\Delta\mathbf{p}_t\|_2^2+\varepsilon}, 0, 1\big)$ denote the closest-point parameter along $\Delta\mathbf{p}_t = \mathbf{p}_t - \mathbf{p}_{t-1}$, with interpolated tool-tip reference $\mathbf{c}_t(\mathbf{x}) = \mathbf{p}_{t-1} + \tau^\star(\mathbf{x})\Delta\mathbf{p}_t$. The local swept cutting-flute SDF is

$$\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t) = \max\Big(\|\mathbf{x}^{xy} - \mathbf{c}_t^{xy}(\mathbf{x})\|_2 - R, \; c_t^z(\mathbf{x}) - x^z, \; x^z - \big(c_t^z(\mathbf{x}) + L_{\text{flute}}\big)\Big). \tag{1}$$

In exact hard Boolean mode, material subtraction updates the IPW via the exact CSG difference:

$$\phi_t^{\text{hard}}(\mathbf{x}) = \max\big(\phi_{t-1}^{\text{hard}}(\mathbf{x}), \; -\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t)\big). \tag{2}$$

Because $\max(a, b)$ has zero derivative with respect to inactive arguments and a discontinuous subgradient along $a = b$, we pair Eq. (2) with a numerically stabilized smooth log-sum-exp relaxation controlled by inverse-temperature stiffness $k > 0$:

$$\begin{aligned}\phi_t^{(k)}(\mathbf{x}) &= \max_k\big(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x})\big) \\ &= m + \frac{1}{k}\log\Big(e^{k(\phi_{t-1}^{(k)}(\mathbf{x})-m)} + e^{k(-\psi_{\text{tool}}(\mathbf{x})-m)}\Big),\end{aligned} \tag{3}$$

where $m = \max(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$. Smooth voxel occupancy is mapped via the logistic sigmoid $\sigma_k(\phi) = (1 + e^{k\phi})^{-1} \in (0, 1)$, which converges pointwise to the hard indicator $\mathbb{I}[\phi < 0]$ as $k \to \infty$ while providing dense, analytically differentiable reverse-mode sensitivities $\partial\mathcal{L}_k/\partial\mathbf{p}_t$ across the $\mathcal{O}(1/k)$ boundary layer.

B. Straight-Through Estimator and Stiffness Continuation

Optimizing $\mathcal{L}_k$ at a fixed low stiffness (e.g., $k \le 20$) suffers from severe surrogate occupancy drift: the $\frac{\log 2}{k}$ additive shift in Eq. (3) accumulates over overlapping passes, inflating soft stock erosion relative to ϕ_T^{hard}. Conversely, initializing directly at $k \ge 100$ concentrates gradients onto a razor-thin shell. We resolve this bias-variance trade-off via a Straight-Through Estimator (STE) with linear stiffness continuation. At optimization step $n \in \{0, \dots, N-1\}$, we schedule $k(n) = k_{\min} + \frac{n}{N-1}(k_{\max} - k_{\min})$ $(25 \to 90)$ and define the forward-hard / backward-smooth objective:

$$\mathcal{L}_{\text{STE}}(\boldsymbol{\theta}; k(n)) = \mathcal{L}_{\text{hard}}(\boldsymbol{\theta}) + \Big(\mathcal{L}_{k(n)}(\boldsymbol{\theta}) - \text{sg}\big[\mathcal{L}_{k(n)}(\boldsymbol{\theta})\big]\Big), \tag{4}$$

where $\text{sg}[\cdot]$ is the stop-gradient operator. Every candidate $\boldsymbol{\theta}_n$ is scored and checkpointed strictly on exact Boolean $\mathcal{L}_{\text{hard}}(\boldsymbol{\theta}_n)$ and hard geometric/safety feasibility, while reverse-mode Taichi kernels supply $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{k(n)}(\boldsymbol{\theta}_n)$ at the cost of one additional hard forward pass (< 8 ms on GPU).

C. Blended Acceleration-Bounded CNC Kinematics

To eliminate the mismatch between idealized constant-speed models and modern CNC lookahead controllers, we formulate a differentiable acceleration-bounded corner-blending timing law (BLENDED_ACCEL_BOUNDED) aligned with RS274 G64 P0.05 and Haas NGC G187 P2 E0.05. Let segment t have length $\ell_t = \|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2$, unit tangent $\hat{\mathbf{d}}_t = (\mathbf{p}_t - \mathbf{p}_{t-1})/(\ell_t + \varepsilon)$, and commanded speed $v_t = f_t$ for cutting moves (G1) or $v_t = v_{\text{rapid}}$ for non-cutting rapids (G0). At interior junction t between consecutive cutting moves, the turning angle α_t satisfies $\cos\alpha_t = \hat{\mathbf{d}}_t \cdot \hat{\mathbf{d}}_{t+1}$, yielding the smooth half-angle sine

$s_t = \sqrt{\frac{1}{2}(1-\cos\alpha_t)+\varepsilon}$ and blended junction speed

$$v_{\text{junc},t} = \begin{cases} \min(v_t, v_{t+1})(1-\kappa_c s_t), & \text{G1–G1 junction,} \\ 0, & \text{otherwise,} \end{cases} \tag{5}$$

with blending factor $\kappa_c = 0.85$ calibrated to a 0.05 mm corner-rounding tolerance under machine acceleration limit $a_{\max}$. With entry speed $v_{\text{in},t} = \min(v_t, v_{\text{junc},t-1})$ and exit speed $v_{\text{out},t} = \min(v_t, v_{\text{junc},t})$ bounded by reachability over ℓ_t, the segment duration and total cycle time are

$$\begin{aligned} \tau_t(\mathbf{p}, \mathbf{f}) &= \frac{\ell_t}{v_t} + \frac{(v_t - v_{\text{in},t})^2 + (v_t - v_{\text{out},t})^2}{2\, a_{\max}\, v_t}, \\ T_{\text{cycle}} &= \sum_{t=1}^{T} \tau_t(\mathbf{p}, \mathbf{f}). \end{aligned} \tag{6}$$

Crucially, τ_t is jointly differentiable with respect to both waypoint positions $\mathbf{p}$ (via ℓ_t and junction deflection $\cos\alpha_t$) and per-segment feedrates f_t.

D. Volumetric Peak MRR and Machining Safety Barriers

Increasing feedrate f_t reduces τ_t in Eq. (6), but unchecked acceleration in full-width slotting or corner plunges exceeds cutter flute evacuation and spindle torque limits [1]. At each cutting step t, our simulator integrates the pre-cut swept chip volume directly from the evolving IPW field ϕ_{t-1}:

$$\begin{aligned} \Delta V_t &= \Delta x^3 \sum_{\mathbf{x}\in\Omega} \sigma_k\big(-\phi_{t-1}(\mathbf{x})\big)\, \sigma_k\big(-\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t)\big), \\ \text{MRR}_t &= \frac{\Delta V_t}{\tau_t}. \end{aligned} \tag{7}$$

We enforce a rated volumetric removal limit $\text{MRR}_{\max}$ via a horizon-normalized quadratic overload barrier $\mathcal{B}_{\text{MRR}} = w_{\text{MRR}} \sum_{t=1}^{T} \max\big(0, \frac{\text{MRR}_t}{\text{MRR}_{\max}} - 1\big)^2$. Under this coupling, air-cutting and light radial-skimming passes ($\Delta V_t \approx 0$) receive pure cycle-time gradients that push $f_t \to f_{\max}$, whereas heavy roughing passes automatically throttle f_t or shift waypoints laterally to shave ΔV_t.

Given target blueprint SDF $\phi_{\text{target}}(\mathbf{x})$, final geometric accuracy combines overcut gouge $V_{\text{gouge}} = \Delta x^3 \sum_{\mathbf{x}} \mathbb{I}[\phi_{\text{target}}(\mathbf{x}) < 0]\, \sigma_k(\phi_T(\mathbf{x}))$ and uncut residual $V_{\text{res}} = \Delta x^3 \sum_{\mathbf{x}} \mathbb{I}[\phi_{\text{target}}(\mathbf{x}) > 0]\, \sigma_k(-\phi_T(\mathbf{x}))$, augmented by three analytical machine-protection barriers: (1) an exact continuous 2D point-to-segment wall-clearance barrier $\mathcal{B}_{\text{wall}}$ preventing sub-voxel finishing-wall nicks, (2) a cylindrical tool-holder collision barrier $\mathcal{B}_{\text{holder}}$ over $z \in [c_t^z + L_{\text{clear}}, c_t^z + L_{\text{clear}} + L_{\text{holder}}]$ of radius R_{holder} against ϕ_{t-1}, and (3) a G0 rapid-stock intersection barrier $\mathcal{B}_{\text{rapid}}$ guaranteeing zero rapid contact with unmachined stock. The composite loss is

$$\begin{aligned} \mathcal{L} = w_T \frac{T_{\text{cycle}}}{T_{\text{nom}}} + w_g \bar{V}_{\text{gouge}}^2 + w_r \bar{V}_{\text{res}}^2 \\ + \mathcal{B}_{\text{MRR}} + \mathcal{B}_{\text{wall}} + \mathcal{B}_{\text{holder}} + \mathcal{B}_{\text{rapid}}, \end{aligned} \tag{8}$$

with candidate hard feasibility requiring simultaneously zero holder collision, zero rapid stock contact, and target gouge/residual below strict volumetric tolerances.

IV. Trajectory and Policy Co-Optimization

A. Drift-Free Waypoint-Position and Feed Co-Parameterization

A critical failure mode in long-horizon machining optimization is cumulative delta drift: if an optimizer perturbs segment displacements $\Delta\mathbf{p}_t = \Delta\bar{\mathbf{p}}_t + \delta\mathbf{p}_t$, waypoint $\mathbf{p}_t = \mathbf{p}_0 + \sum_{j=1}^{t} \Delta\mathbf{p}_j$ accumulates $O(T)$ random-walk variance across $T \in [37, 370]$ segments, causing downstream finishing passes or vertical retracts to collide with pocket walls even under tiny step sizes.

We eliminate cumulative drift via a bounded Waypoint-Position and Log-Feed Co-Parameterization (JointPathFeedSpline). Given nominal CAM waypoints $\{\bar{\mathbf{p}}_i\}_{i=0}^{T}$ and nominal cutting feeds $\{\bar{f}_t\}_{t=1}^{T}$, we hold axial heights $\bar{p}_i^z$ fixed to preserve constant-layer floor flatness and safe G0 clearance planes, and parameterize horizontal waypoints and segment feeds as

$$\begin{aligned} \mathbf{p}_i^{xy}(\boldsymbol{\theta}) &= \bar{\mathbf{p}}_i^{xy} + r_{xy}\, w_i \tanh\big((\mathbf{B}\boldsymbol{\theta}_{\text{pos}})_i\big), \\ f_t(\boldsymbol{\theta}) &= \min\Big(0.995\, v_{\max},\ \bar{f}_t \exp\big(\rho_f \tanh(\theta_{\text{feed},t})\big)\Big), \end{aligned} \tag{9}$$

where $\mathbf{B} \in \mathbb{R}^{(T+1)\times K_p}$ is a clamped cubic B-spline basis ($K_p \le 40$), $r_{xy} = 1.25$ mm bounds maximum lateral excursion, and $\rho_f = 1.25$ scales log-feed adjustments up to the machine feed ceiling $v_{\max} = 20.0$ mm/s. Crucially, $w_i \in [0, 1]$ is an analytical blueprint wall-clearance weight:

$$w_i = \begin{cases} 0, & \text{if } i \text{ is a plunge/retract,} \\ \text{clip}\big(\frac{d_{\text{wall}}(\bar{\mathbf{p}}_i) - \delta_{\text{pin}}}{r_{xy}},\, 0,\, 1\big), & \text{otherwise,} \end{cases} \tag{10}$$

where $d_{\text{wall}}(\bar{\mathbf{p}}_i)$ is the exact Euclidean clearance from the cutter periphery at $\bar{\mathbf{p}}_i$ to any retained pocket wall or island boss ($\delta_{\text{pin}} = 0.08$ mm). Equation (10) automatically pins vertical plunge/retract axes and tangent wall-finishing passes ($w_i = 0$) while granting full $O(1)$-bounded lateral mobility to interior roughing passes.

B. Invariant Segment State and Residual RL Policies

To test whether machining heuristics can generalize across part dimensions without re-solving from scratch, we cast segment-level refinement as a finite-horizon Markov Decision Process with an invariant 18-D per-segment state $\mathbf{s}_t \in \mathbb{R}^{18}$ extracted from a single nominal pass: normalized start/end coordinates in stock-box coordinates (6), unit motion direction and normalized length (4), normalized nominal feed and adjacent junction cosines $(\cos\alpha_{t-1}, \cos\alpha_t)$ (3), binary cutting/plunge/retract flags (2), normalized pre-cut chip volume ΔV_t, nominal $\text{MRR}_t/\text{MRR}_{\max}$, local target SDF clearance, and endpoint mobility $(w_{t-1} + w_t)/2$ (3).

A shared two-layer tanh MLP policy $\boldsymbol{\pi}_\psi(\mathbf{a}_t \mid \mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_\psi(\mathbf{s}_t), \text{diag}(\boldsymbol{\sigma}^2))$ (64–64 hidden units, zero-initialized

output layer so $\boldsymbol{\mu}_{\psi_0}(\mathbf{s}_t) \equiv \mathbf{0}$ starts exactly at the feasible nominal CAM seed) outputs residual waypoint and log-feed actions $\mathbf{a}_t \in [-1, 1]^3$ mapped through Eq. (9) via DirectSegmentActionMap. We compare two controlled policy-learning paradigms sharing identical network architectures and simulator budgets:

1) DiffPolicy-SHAC (First-Order Differentiable RL): Following Short-Horizon Actor-Critic [20], we roll out policy actions $\mathbf{u} = \boldsymbol{\mu}_\psi(\mathbf{S})$ through our STE continuation evaluator (Eq. (4)) and backpropagate analytical simulator gradients $\nabla_{\mathbf{u}}\mathcal{L}_{k(n)}$ directly into ψ via $\nabla_\psi \mathcal{L} = (\partial \boldsymbol{\mu}_\psi / \partial \psi)^\top \mathbf{P} \nabla_{\mathbf{u}} \mathcal{L}_{k(n)}$. Here $\mathbf{P}$ applies decoupled norm preconditioning across spatial XY and log-feed coordinates so stiff wall-barrier gradients never drown out smooth feedrate sensitivities during Adam [9] updates.
2) Hard-PPO (Zero-Order Clipped Policy Gradient): Operating purely on exact hard Boolean rollouts ($\nabla_{\mathbf{u}}\mathcal{L}$ unavailable), we train $\boldsymbol{\pi}_\psi$ alongside a 64–64 ValueCritic $V_\omega(\mathbf{s}_t)$ using Clipped Proximal Policy Optimization ($\epsilon = 0.2$) [15] and Generalized Advantage Estimation ($\gamma = 0.99, \lambda = 0.95$) [14], driven by dense per-segment hard rewards $r_t = -\left(\frac{\tau_t}{T_{\text{nom}}} + w_{\text{MRR}} \max(0, \frac{\text{MRR}_t}{\text{MRR}_{\max}} - 1)^2\right) - \frac{1}{T}\mathcal{C}^{\text{hard}}_{\text{terminal}}$.

V. Experimental Setup

MIT Clean-Room 16-Task Milling Suite. We construct a 100% MIT-licensed benchmark suite seeded by four modular clean-room 2.5D CAM strategy engines—facing_open_boss (zig-zag raster with island contour, 37–39 segments), pocket_island (inward/outward offset rings with corner-dwell cleanup, 92–169 segments), adaptive_channel (engagement-bounded trochoidal clearing around dual islands, 261–370 segments), and contour_profile (outside-in exterior boss profiling with roll-around corner arcs, 156–188 segments)—with 2 development (dev) and 2 held-out test (test) instances per family (16 tasks total). All tasks model Al 6061-T6 billets (30–38 × 24–32 × 8 mm) machined by a 5.0 mm diameter ($R = 2.5$ mm) flat carbide endmill (12 mm flute, 16 mm holder diameter at 14 mm clearance, 8000 RPM, $v_{\max} = 20$ mm/s, $v_{\text{rapid}} = 100$ mm/s, $a_{\max} = 150$ mm/s^2, and rated $\text{MRR}_{\max} \in [95, 120]$ mm^3/s).

Methods and Strict Simulator-Call Parity. Every optimizer is audited by a deterministic ComputeLedger capped at $B = 64$ total simulator forward/backward invocations across 3 random seeds $(0, 1, 2)$ on an NVIDIA RTX 5070 Ti GPU (0.50 mm optimization grid): (0) Nominal-CAM, (1) Naive-Smooth-Adam (fixed $k = 25$), (2) Annealed-Adam ($k : 20 \to 85$ with periodic 4:1 hard checkpointing, yielding 51 gradient updates), (3) STE-Continuation-Adam ($k : 25 \to 90$ with 1:1 per-step hard STE selection, yielding 32 gradient updates), (4) DiffPolicy-SHAC (1st-order RL, 32 STE updates), (5) Hard-PPO (0th-order RL, 8 episodes/batch), (6) Hard-CEM (population 12, elite 25%), and (7) Hard-CMA-ES [6] ($\sigma_0 = 0.22$, population 12).

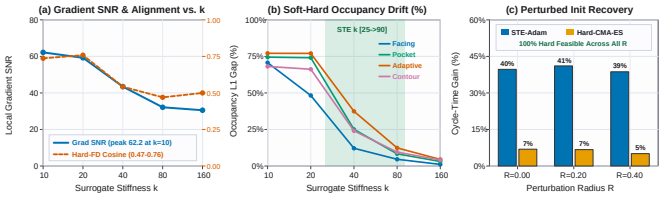


Fig. 2. Exp1: Surrogate gradient conditioning and perturbed recovery. (a) Analytical gradient SNR ($62.2 \to 30.5$, blue) and hard finite-difference cosine ($0.76 \to 0.47$, orange right axis) across k. (b) Soft-to-hard occupancy L_1 bias collapses from 72.6% ($k = 10$) to 8.7% ($k = 80$) and 3.2% ($k = 160$) across our STE continuation window. (c) From perturbed initializations $R \in \{0.00, 0.20, 0.40\}$, STE-Continuation-Adam achieves 100% feasibility and 38.7%–41.1% speedups vs. 5.1%–6.9% for Hard-CMA-ES.

6-Controller G-Code, CAMotics STL, and 3D Simulators. Held-out test solutions are compiled across six clean-room industrial post-processors—NIST RS274 (G64 P0.05), Haas Mini Mill Next-Gen Control (G54, G187 P2 E0.05, O01001–O01008), FANUC 30i/31i (G05.1 Q1), FANUC Robodrill, Makino Pro 5/6 (Super GI), and Mazak Smooth/Matrix (G61.1 EIA/ISO)—generating 48 machine programs alongside native .rustcnc Manufacture Workspace containers, .camotics project bundles, and an interactive WebGL 3D stock-and-toolpath viewer. Every test part is audited at 2× finer spatial resolution (0.25 mm grid) by both a standalone 0.125 mm-step swept-SDF posted_evaluator and containerized CAMotics v1.2.0 [3].

VI. Results and Analysis

A. Exp1: Surrogate Conditioning and Perturbed Reachability

Figure 2 characterizes the numerical mechanics of our smooth CSG relaxation across stiffness $k \in \{10, 20, 40, 80, 160\}$ on the four development families. In Fig. 2(a), local gradient signal-to-noise ratio (SNR) and directional cosine similarity against central finite differences of exact $\mathcal{L}_{\text{hard}}$ peak at $k \in \{10, 20\}$ (SNR = **62.19**–**59.24**, $\cos\theta$ = **0.737**–**0.758**), declining gradually to SNR $= 43.47, 32.14, 30.53$ ($\cos\theta = 0.543, 0.470, 0.500$) at $k \in \{40, 80, 160\}$ as active boundary shells thin across 37–370 multi-pass segments. Conversely, Fig. 2(b) reveals why low-stiffness optimization fails without continuation: the L_1 volumetric occupancy gap between soft and hard IPW states is **72.6**% at $k = 10$ and **66.4**% at $k = 20$, but drops sharply to **24.7**% at $k = 40$, **8.7**% at $k = 80$, and **3.2**% at $k = 160$. Sweeping $k : 25 \to 90$ under STE captures wide-basin SNR early and sharp geometric fidelity late.

In Fig. 2(c), we perturb initial parameters uniformly within $\|\boldsymbol{\theta}_0\|_\infty \le R$ for $R \in \{0.00, 0.20, 0.40\}$ (48 simulator calls). STE-Continuation-Adam maintains 100% hard feasibility across all R and achieves **39.7**%, **41.1**%, and

38.7% cycle-time reductions—5.8× to 7.6× the gain of zero-order Hard-CMA-ES (6.9%, 6.7%, 5.1%).

B. Exp2: Single-Task Trajectory and RL Benchmark

Table I and Fig. 3 report our main 16-task × 8-method benchmark under strict $B = 64$ simulator-call parity across 3 seeds (100% hard feasibility across all methods). Both smooth-to-hard continuation schedules substantially outperform fixed-stiffness and zero-order baselines, cutting mean cycle time from 122.20 s to **73.91** s (STE-Continuation-Adam, $-\mathbf{42.0} \pm \mathbf{13.5}$%) and **73.74** s (Annealed-Adam, $-\mathbf{43.0} \pm \mathbf{15.7}$%)—more than 10 percentage points ahead of fixed-k Naive-Smooth-Adam (−31.9%, 83.44 s) and nearly 4× the speedup of evolutionary search (Hard-CMA-ES −11.3%, Hard-CEM −9.3%), which stall on 156–370 segment Contour and Adaptive programs. Comparing our two continuation schedules under identical simulator calls exposes a clear compute-allocation trade-off: 1:1 per-step STE (STE-Continuation-Adam, 32 gradient updates) achieves the highest speedups on engagement-varying Facing ($-\mathbf{59.7} \pm \mathbf{1.2}$%) and long-horizon trochoidal Adaptive clearing ($-\mathbf{37.3} \pm \mathbf{7.2}$% vs. −33.0% for Annealed-Adam) while halving seed variance on walled Pocket roughing (±9.7% vs. ±19.2%), whereas 4:1 periodic hard checkpointing (Annealed-Adam, 51 gradient updates) converts extra gradient steps into a lead on exterior Contour profiling ($-\mathbf{52.4} \pm \mathbf{3.4}$%).

Comparing our controlled RL arms directly answers the first- vs. zero-order policy learning question: under identical 64–64 MLP architectures and a 64-call simulator budget, first-order DiffPolicy-SHAC achieves $-\mathbf{37.0} \pm \mathbf{12.7}$% mean cycle-time reduction (81.67 s), doubling (2.0×) the speedup of zero-order Hard-PPO ($-\mathbf{18.5} \pm \mathbf{6.6}$%, 102.26 s) and more than tripling Hard-CMA-ES (−11.3%). Because Hard-PPO must estimate credit assignment across up to 370 sequential segments from Gaussian action noise, 64 rollouts provide only 8 policy updates. By contrast, DiffPolicy-SHAC backpropagates exact per-segment analytical sensitivities on every step—and notably achieves the highest Pocket speedup of any method ($-\mathbf{32.1} \pm \mathbf{12.4}$% vs. −26.9% for spline-based STE) because its per-segment DirectSegmentActionMap accelerates individual corner-cleanup moves without B-spline basis smoothing across 92–169 pocket segments.

C. Exp3: Multi-Task Policy Transfer and Physical Ablations

In Exp3 (Table II top, Fig. 4a), we train multi-task DiffPolicy-SHAC and Hard-PPO policies across the 8 dev tasks and evaluate generalization on the 8 unseen test tasks:

- Zero-Shot vs. Overload Trade-Off: Across heterogeneous toolpath topologies (37-segment raster facing to 370-segment trochoidal loops), zero-shot DiffPolicy-SHAC transfers strongly within low-horizon Facing (+**35.4**% in < 1.5 ms) while remaining conservative near nominal feeds on unseen multi-ring/trochoidal geometries (8.3% overall, 16.9% MRR overload). Conversely, Hard-PPO-ZeroShot applies an aggressive uniform feed boost (26.1% speedup) that violates soft chip-load ratings with a +**24.5**% Peak MRR overload (133.2 mm^3/s).
- 16-Call Few-Shot Adaptation: Given just 16 test-time simulator calls, analytical gradients boost DiffPolicy-SHAC-FewShot16 by +19.4 points to $\mathbf{27.7} \pm \mathbf{15.8}$% while cutting MRR overload nearly 4× (16.9% → **4.4**%), surpassing Hard-PPO-FewShot16 (23.4%, which degrades from zero-shot under noisy 4-batch exploration), while from-scratch Scratch-STE-Adam-16 reaches $\mathbf{40.1} \pm \mathbf{7.4}$% (4.0% overload).

Our 5-arm ablation on the 8 held-out test tasks (Table II bottom, Fig. 4b) confirms three core physical hypotheses: (1) Path-Only-Frozen-Feed yields only $\mathbf{0.2} \pm \mathbf{0.4}$% speedup because clean-room stepovers and blueprint walls lock pass topology under constant feedrates; (2) under 32 STE updates ($B = 64$), joint XY-and-feed co-optimization (Full-Joint-Blended-STE, 37.0%) matches Feed-Only-Frozen-Path (39.5%) and Exact-Stop-Surrogate (39.2%) within variance while reducing Peak MRR overload by **29**% relative to frozen-path feed scheduling (**5.2**% vs. 7.3%, 114.9 vs. 118.7 mm^3/s) by laterally easing roughing engagement; and (3) setting $w_{\mathrm{MRR}} = 0$ (No-MRR-Barrier) inflates apparent gain to 66.2% by commanding $v_{\max}$ through full-width slotting cuts, driving mean peak MRR to **262.6** mm^3/s (+**136.7**% overload).

D. Exp4: Multi-Controller NC, CAMotics STL, and Simulator Audit

To verify that optimization on a 0.50 mm grid introduces zero discretization or post-processing regressions, Table III audits all 8 held-out test parts compiled into Haas Mini Mill Next-Gen Control programs (O01001–O01008, G54, G187 P2 E0.05) and across all 6 industrial CNC dialects (48 programs total). Mean test cycle time drops from **128.80** s to **82.13** s (−**36.2**% total machining time, −**39.5**% unweighted mean per part, up to −**60.9**% on face-test-01). Every single program across all 48/48 controller outputs passes our 0.25 mm-voxel, 0.125 mm-step posted_evaluator geometric and holder/rapid safety screen with exactly **0.00** mm^3 target gouge and **0.00** mm^3 uncut residual. Independently, containerized CAMotics v1.2.0 reconstructs **8/8** closed, manifold, watertight STL solids (72,576–129,296 facets) whose enclosed volumes and facet counts match Nominal-CAM identically ($\Delta V = \mathbf{0.000}$% on every part), all inspectable interactively alongside 3D toolpaths in our .rustcnc, .camotics, and WebGL simulator bundles.

VII. Limitations and Conclusion

Limitations. First, our current parameterization locks axial coordinates $\bar{p}_i^z$ to canonical 2.5D cutting layers;

TABLE I

Exp2 Main Benchmark across 16 Milling Tasks (4 Families × 4 Geometries × 3 Seeds, Strict Budget $B = 64$ Simulator Calls). Cycle-time reductions ($-\Delta T_{\text{cycle}}$, %) are mean ± std relative to Nominal-CAM; all methods achieve 100% hard Boolean feasibility under blended acceleration kinematics.

Method	Optimization Class	Facing (%)	Pocket (%)	Adaptive (%)	Contour (%)	Overall $-\Delta T$ (%)	Mean T (s)	Peak MRR (mm^3/s)
Nominal-CAM	Clean-Room CAM Seed	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	122.20	75.4
Hard-CEM [13]	0th-Order Elite Search	15.6 ± 2.6	6.3 ± 3.4	6.1 ± 1.2	9.3 ± 2.0	9.3 ± 4.5	112.17	88.8
Hard-CMA-ES [6]	0th-Order Covariance ES	22.3 ± 2.1	6.3 ± 3.5	6.0 ± 1.8	10.6 ± 3.4	11.3 ± 7.2	110.62	95.9
Hard-PPO [15]	0th-Order Clipped RL	21.0 ± 3.4	20.5 ± 5.6	10.1 ± 4.4	22.2 ± 4.5	18.5 ± 6.6	102.26	93.9
Naive-Smooth-Adam	1st-Order Fixed $k = 25$	33.6 ± 8.1	24.2 ± 14.0	33.3 ± 4.4	36.4 ± 5.2	31.9 ± 9.9	83.44	105.9
DiffPolicy-SHAC [20]	1st-Order Analytic RL	53.6 ± 2.8	$\mathbf{32.1 \pm 12.4}$	24.6 ± 4.2	37.7 ± 2.6	37.0 ± 12.7	81.67	107.4
Annealed-Adam	1st-Order 4:1 k-Sched	57.4 ± 3.1	29.2 ± 19.2	33.0 ± 3.5	$\mathbf{52.4 \pm 3.4}$	$\mathbf{43.0 \pm 15.7}$	**73.74**	109.7
STE-Continuation-Adam	1st-Order 1:1 STE (Ours)	$\mathbf{59.7 \pm 1.2}$	26.9 ± 9.7	$\mathbf{37.3 \pm 7.2}$	44.1 ± 3.7	$\mathbf{42.0 \pm 13.5}$	73.91	113.3

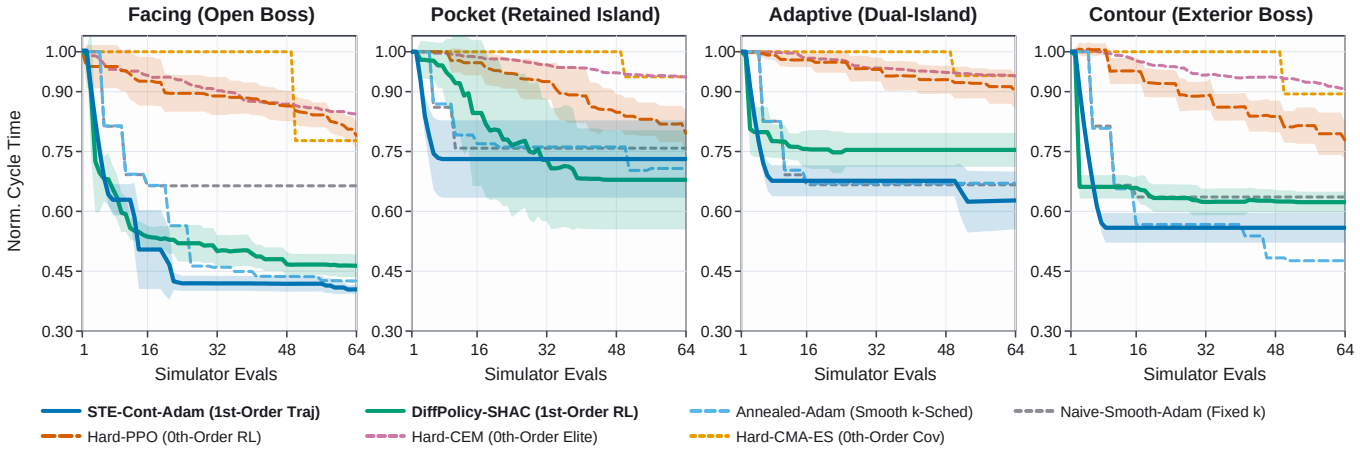


Fig. 3. Exp2 sample-efficiency convergence across clean-room CAM families ($B = 64$ simulator calls, mean ± 1 SD shaded). First-order methods (STE-Continuation-Adam, Annealed-Adam, DiffPolicy-SHAC) achieve steep cycle-time reductions within the first 16–24 calls by exploiting exact per-segment $\partial T_{\text{cycle}}/\partial f_t$ and $\partial \Delta V_t/\partial \mathbf{p}_t$ sensitivities, whereas zero-order RL (Hard-PPO) and evolutionary search (Hard-CMA-ES, Hard-CEM) progress slowly over 100–370 segment horizons.

TABLE II

Top: Exp3 Multi-Task Policy Transfer (8 Dev → 8 Unseen Test Tasks, 3 Seeds). Bottom: Exp2 5-Arm Physical & Kinematic Ablations (8 Test Tasks, $B = 64$). All arms achieve 100% geometric feasibility; removing the MRR barrier triggers +136.7% cutter overload.

Regime / Ablation Arm	Calls	$-\Delta T_{\text{cycle}}$ (%)	Peak MRR	Overload
Exp3: Unseen Test Policy Generalization				
Hard-PPO-ZeroShot	1	26.1 ± 9.5	133.2	24.5%
DiffPolicy-SHAC-ZeroShot	1	8.3 ± 17.9	124.9	16.9%
Hard-PPO-FewShot16	16	23.4 ± 9.5	104.9	2.7%
DiffPolicy-SHAC-FewShot16	16	$\mathbf{27.7 \pm 15.8}$	111.4	4.4%
Scratch-STE-Adam-16	16	$\mathbf{40.1 \pm 7.4}$	114.4	4.0%
Exp2-Abl: Physical & Kinematic Ablations ($B = 64$)				
Path-Only-Frozen-Feed	64	0.2 ± 0.4	82.1	0.0%
Full-Joint-Blended-STE	64	37.0 ± 17.9	114.9	**5.2**%
Exact-Stop-Surrogate	64	39.2 ± 18.2	109.6	2.3%
Feed-Only-Frozen-Path	64	$\mathbf{39.5 \pm 13.7}$	118.7	7.3%
No-MRR-Barrier ($w_{\text{MRR}} = 0$)	64	66.2 ± 3.5	**262.6**	**136.7**%

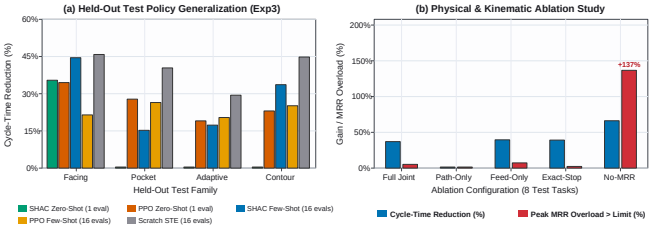


Fig. 4. Exp3 policy generalization and physical ablations. (a) Held-out test cycle-time reductions across families for zero-shot (1 call) and few-shot (16 calls) transfer. (b) Ablating feed optimization (Path-Only) collapses gains to 0.2%, whereas ablating the volumetric MRR barrier (No-MRR) triggers +136.7% peak cutter overload.

extending wall-clearance mobility weights to simultaneous 3-axis ball-endmill sculptured surfacing and 5-axis tool-orientation smoothing is a natural next step. Second, while our volumetric Peak MRR barrier prevents steady-state chip-load and torque overload, it does not model frequency-domain regenerative chatter lobes [1] or thin-wall deflection. Third, although DiffPolicy-SHAC achieves 2.0× single-task sample efficiency over Hard-PPO and rapid 16-call adaptation across unseen geometries, zero-shot generalization across wildly varying horizon lengths (37–370 segments) motivates graph or transformer tool-path backbones, and topology-changing pass insertion still relies on upstream geometric CAM seeding. Finally, while our Haas Mini Mill NGC shop packets (O01001–O01008) are certified in CAMotics v1.2.0 and high-resolution swept-SDF simulation, instrumented physical Al 6061-T6 cutting trials measuring spindle load and profilometry roughness constitute our immediate hardware milestone.

Conclusion. We presented DiffCAM, an MIT-licensed dual-mode differentiable 3-axis milling simulator, clean-

TABLE III

Exp4 Independent High-Resolution Verification, 6-Controller Audit, and Haas Mini Mill NGC Shop Packet Summary Across All 8 Held-Out Test Parts. Audited on posted G-code across 6 dialects (RS274, Haas NGC G187 P2 E0.05, FANUC, Robodrill, Makino, Mazak; 48/48 pass) via a 0.125 mm-step posted_evaluator (0.25 mm voxels) and containerized CAMotics v1.2.0 (0.25 mm octree STL reconstruction).

Haas Prog.	Case ID	CAM Family	Stock (mm)	T_{nom} (s)	T_{DiffCAM} (s)	$-\Delta T$ (%)	Gouge (mm^3)	Res. (mm^3)	6-Dialect	CAMotics STL	STL Vol. (mm^3)
O01001	face-test-00	Facing Boss	34×24×8	79.92	32.85	58.9%	0.00	0.00	6/6 Pass	Watertight (74.9k△)	4245.9
O01002	face-test-01	Facing Boss	32×24×8	77.68	30.34	60.9%	0.00	0.00	6/6 Pass	Watertight (72.6k△)	4370.6
O01003	pocket-test-00	Pocket Island	32×28×8	117.24	96.26	17.9%	0.00	0.00	6/6 Pass	Watertight (107.9k△)	5034.5
O01004	pocket-test-01	Pocket Island	34×26×8	89.94	62.01	31.0%	0.00	0.00	6/6 Pass	Watertight (102.1k△)	5514.2
O01005	adapt-test-00	Adaptive Trochoid	38×30×8	188.64	123.36	34.6%	0.00	0.00	6/6 Pass	Watertight (129.3k△)	6675.4
O01006	adapt-test-01	Adaptive Trochoid	36×32×8	280.60	198.18	29.4%	0.00	0.00	6/6 Pass	Watertight (127.1k△)	7060.9
O01007	contour-test-00	Contour Boss	32×26×8	107.81	59.82	44.5%	0.00	0.00	6/6 Pass	Watertight (76.1k△)	4143.0
O01008	contour-test-01	Contour Boss	32×24×8	88.60	54.19	38.8%	0.00	0.00	6/6 Pass	Watertight (73.2k△)	4204.8
Mean / Pass	8 Held-Out Test Parts (36.2% Agg. Time Saved)			**128.80**	**82.13**	**39.5**%	**0.00**	**0.00**	**48/48**	**8/8** Watertight	$\Delta V = 0.000\%$

room CAM suite, and co-optimization framework that reconciles discontinuous Boolean CSG geometry with smooth analytical policy and trajectory gradients. By combining Straight-Through Estimator stiffness continuation, controller-matched corner-blending kinematics, volumetric Peak MRR barriers, and drift-free JointPathFeedSpline parameterizations, DiffCAM achieves 42%–43% mean cycle-time reductions at 100% feasibility, demonstrates a 2.0× sample-efficiency advantage for first-order differentiable RL over zero-order RL, and delivers certified watertight CAMotics STL solids, interactive 3D simulator workspaces, and shop-ready 6-controller G-code.

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